

Rithwick Gurram

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TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, Go, SQL

CS Fundamentals: Data Structures & Algorithms, OOP, Database Systems (SQL, ACID, Normalization), Operating Systems, Computer Networks

Backend: Node.js, Express.js, REST API Design, JWT Authentication & Authorization, Layered Architecture, Redis Caching, Rate Limiting, Connection Pooling, Load Testing (k6)

Databases: PostgreSQL, Prisma ORM, pgx, Database Design, Redis

Frontend & Testing: React, Vite, HTML/CSS, Vitest

DevOps & Cloud: Git, Docker, Docker Compose, GitHub Actions (CI/CD), Google Cloud Run, Cloud SQL, Cloud Storage, Memorystore, Artifact Registry, Postman

PROJECTS

URL Shortener

github.com/Dunkrick/url-shortener

- Built and deployed a stateless Go URL-shortening service on Google Cloud Run, with Cloud SQL PostgreSQL as the source of truth (deterministic Base62 codes, unique constraint with `ON CONFLICT`) and Memorystore Redis for cache-aside redirects.
- Root-caused a `conn busy` failure under k6 load to a shared `pgx.Conn` and fixed it with `pgxpool.Pool`, reaching 135.5 req/s with 0% errors on the redirect path.
- Added a Redis cache-aside layer (1-hour TTL, PostgreSQL fallback on miss), improving throughput 13.8% (135.5 to 154.3 req/s) and p95 latency 23.6% (99.5 to 76.1 ms) under the same 10-VU k6 workload.
- Implemented Redis-backed fixed-window rate limiting (atomic counters, 100 creates/min per client IP, HTTP 429) so the limit is shared across Cloud Run instances; verified locally and in production, and documented the architecture and trade-offs in a system-design doc.

Nook

github.com/Dunkrick/nook | Live Demo

- Built a full-stack TypeScript workspace app with JWT authentication, workspace- and artifact-level ownership checks, and a layered route → service → persistence architecture; modeled typed artifacts (text, link, image) with JSON content and persisted (x, y, zIndex) positions in PostgreSQL.
- Built a private image pipeline on Google Cloud Storage behind an `ObjectStorage` abstraction: authenticated uploads (JPEG/PNG/WebP, 10 MB limit), per-user object keys, and ownership-checked retrieval returning short-lived signed URLs as JSON, which fixed a cross-origin redirect (CORS) failure.
- Set up CI/CD with Docker and GitHub Actions (type-check, build, Docker build verification, Vitest backend and frontend tests) and automated Cloud Run deployment with a `/health` check; documented architecture and deployment.

OPEN SOURCE

meow-starter - test: cover mobile module filtering

meow-starter PR #29

- Opened an upstream PR adding regression tests for `mountModules()` on mobile (desktop-only module skipped, mobile-compatible module loaded); full test suite passes (103 tests). PR open.

EDUCATION

Indian Institute of Information Technology Design and Manufacturing, Kancheepuram 2023 - 2027

Bachelor of Technology in Computer Science & Engineering

CGPA: 7.27

CERTIFICATIONS

SQL Intermediate - HackerRank (Mar 2025)

Google AI Essentials (Jan 2026)